

Understanding User Perspective in a University Setting to Improve Biometric Authentication Adoption

Aubrey Patrick
Andrew Burris
Naheem Noah
Sanchari Das
University of Denver
Denver, Colorado, USA

ABSTRACT

Personal devices have become equipped with more sophisticated security methods through biometric authentication, which does not require users to memorize passwords and is inherently unique to the user. However, the nature of biometric inputs can lead to privacy and security concerns. Thus, evaluating the user perspective and finding the balance between ease of use, security, and privacy is imperative. To this aid, we conducted a survey-based study collecting data from 38 university students to see how prevalent these security and usability issues were in the everyday usage of personal devices. The survey assessed the user's technical experience and then asked general device security questions and those specific to the security, privacy, and usability of biometric authentication if they had ever used it. Our results showed that usability is not a significant problem for most users in a university setting and that ease of use is the primary motivator for utilizing this authentication method; however, some users expressed concerns about losing access to the accounts or having incorrect access to their accounts. Thus, we propose design alterations and security measures to address these issues and the concerns about biometric authentication's security implications.

CCS CONCEPTS

• **Security and privacy** → **Human and societal aspects of security and privacy; Usability in security and privacy; • Human-centered computing** → **Human computer interaction (HCI); Empirical studies in HCI.**

KEYWORDS

Biometric Authentication, Usability, Privacy and Security.

ACM Reference Format:

Aubrey Patrick, Andrew Burris, Naheem Noah, and Sanchari Das. 2022. Understanding User Perspective in a University Setting to Improve Biometric Authentication Adoption. In *Proceedings of Ninth Mexican Conference on Human-Computer Interaction (MexIHC '22)*. ACM MexIHC '22, Ninth

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MexIHC '22, November 17-18, 2022, Online Conference

© 2022 Association for Computing Machinery.
ACM ISBN 978-1-4503-XXXX-X/18/06...\$15.00
<https://doi.org/10.1145/3565494.3565498>

Mexican Conference on Human-Computer Interaction, November 17-18, 2022 - Online Conference, 10 pages. <https://doi.org/10.1145/3565494.3565498>

1 INTRODUCTION

Biometric authentication is the process of authenticating the user profile and account identification based on biological and behavioral characteristics, such as face or fingerprint, as a security measure. As more sophisticated biometric authentication technologies have become available to users, and the algorithms have continuously been improved, increased adoption of this authentication method has been recorded [3]. Although biometric authentication prevents users from going through manual password entry, it poses a new set of security risks and usability challenges. Specifically, facial recognition and touch identification, among the most commonly adopted biometric authentication methods [16, 32, 34], collect sensitive data at the expense of a less complicated user authentication process.

Experts in technological privacy and security have argued about the concerns of the authentication process from a security perspective in addition to the perceived usability of the system [24, 42]. Looking at biometric authentication from a security perspective, it is something that experts in technological privacy and security have argued to be equally important as the usability of the systems themselves [24, 42]. With more advanced systems and higher adoption of facial recognition authentication in recent years [3, 7, 9, 13].

Our study aims to evaluate these perceptions once more in order to gain a new perspective from the user side in a university setting. In order to understand users' opinions on the usability and security of biometric authentication, we surveyed 38 students in a university setting. Overall, 87.5% of the respondents had used or were currently using biometric authentication on their devices. We found that users felt that the use of biometric authentication was more secure than a traditional pin. However, about 31% of the respondents felt concerned about someone observing them unlocking their device with their PIN. Our results create the scope of design architecture modification to create a balance of security and usability for biometric authentication. We consider this work foundational for future studies, where we plan to conduct longitudinal studies to understand user perceptions and create robust authentication schemes.

2 RELATED WORK

User authentication is verifying a user's identity attempting to gain access to a network or computing resource [10, 17, 38]. Harbach

et al. found that users spend up to 9% of their screen time dealing with unlocking their devices. This study was conducted shortly after implementing biometric unlocking technology such as Touch ID. Therefore, PIN use served as the primary method of device protection for the users studied in this work. Additionally, it was found that 25.5% of the users studied showed a preference for an easier way to unlock their devices, and 48.8% agreed that biometric authentication on their mobile phone could be annoying [26]. Authentication can take several forms on personal devices, such as a traditional PIN, connecting node pattern, or biometric authentication [6, 12, 14, 18, 30]. For biometric authentication, the user's biological features are input to verify identity rather than a specific personal identification number.

A common misconception about biometric authentication security is the claim that this mode of authentication is not susceptible to password hacking by attackers, which prior work has shown to be untrue. According to a technical study by O'Gorman comparing passwords, tokens, and biometrics for user authentication, "matching precision for a biometric is even lower than its sampling position" [38]. This means it is possible to use machine authentication to hack biometric data to remotely gain authentication into a system, similar to how a PIN would be hacked. Prior research into security concerns with biometric authentication focus on the risk factor of biometric data being stolen because of the sensitivity of the data. The discussion centers around the perceived security of biometrics authentication because of the unique data points collected from an individual. Once the data is compromised or stolen by an attacker, the user cannot modify their data to respond to the data breach. This is highly concerning from a security standpoint [8].

Additionally, it is often claimed that the usability of biometrics is at its best. In this study, we specifically focused on biometric means of authentication rather than a PIN since the technology has vastly improved, and a comprehensive level of adoption has been achieved in recent times. We introduced a survey-based study to evaluate similar perceptions of device protection specifically targeted around biometrics. Some of the main usability concerns noted about biometric authentication systems from previous studies have been obstructions and errors when scanning. For example, having dirty hands can impact the verification of a fingerprint, or wearing sunglasses or makeup could impact facial recognition [11, 14, 35]. Another highly researched topic from a technical standpoint is the usability of facial recognition systems depending on skin color and gender. It has been proven that these systems have a significantly high error rate for individuals with darker skin and individuals who are not male [23].

A study conducted by Wolf et al. follows a similar basis as our study by comparing user perception of biometric authentication security among experts and regular users [42]. The study focused more on Touch ID other than the other available biometric authentication methods and found that regular users find it more beneficial to use the Touch ID; however, experts have expressed some concerns. Past academic research has critiqued facial recognition software as less effective at measuring specific physical characteristics. Cook et al., for example, tested eleven commercial face recognition biometric systems with a diverse set of subjects, in which lower skin reflectance was associated with lower efficiency and accuracy of the systems [7]. Our research incorporates users'

perspectives on the usability of biometric technology as a method of securing their device(s) and what factors are perceived to impact this. Therefore, this study mainly focuses on the users' perception of security versus usability and how this impacts the adoption of any lock on their device.

3 METHOD

We started by looking through several different techniques and user studies on biometric authentication to form a focus for our research. The technical study mentioned above by Cook et al. motivated the usability side of our research focus because it looked at the relationship between certain facial features and the effectiveness of facial recognition systems. We also researched the most common biometric authentication methods used today to broaden our analysis to more than just facial recognition. Lastly, we looked into several different user studies focused on biometric authentication. We found a study done by Wolf et al. that was a study conducted to compare the perceptions of biometric authentication between experts and non-experts when it came to data security. This allowed us to come up with the focus of our research on users' perceptions of the usability and security of several different biometric authentication methods. We also aimed to group our respondents into expert and non-expert categories as a means of comparison.

We commenced with a survey in order to collect user data on their current device security practices for university students ($N = 38$). The survey questions were compiled from previous studies on users' perceptions of biometric usability. The survey questions were adapted from several prior works, one Bhagavatula et al. surrounding why users chose biometric authentication and what issues they faced with the systems, if any, were included [3]. Another set of questions adopted from the study by Heckle et al. about whether users think there are better ways to have more robust security other than biometrics was included [27]. Lastly, several questions from the study conducted by Hardback et al. surrounding general device security practices and users' perceptions of ease of unlocking their devices were included [25]. These questions aimed to understand how users perceived biometric authentication's usability and security. Details of the questions are provided in the appendix.

Additionally, we included a section of the survey on the user's technical experience to gauge how much experience users had with topics in cybersecurity in general. These questions included topics on data security and whether they had ever taken classes on it, whether it was a part of their job, whether they had completed specific security tasks such as downloading a virus scanner or implementing a firewall, etc. These questions were taken from Rajivan et al.'s work [39]. Finally, the survey ended with a few demographic questions focused on age, gender, and race, as well as whether the user perceived their physical appearance to impact the usability of specific biometric authentication systems. In conjunction with the information from the technical study, this can give developers an idea of how to continue improving facial recognition algorithms to increase the system's usability and equity among diverse groups of individuals. Additionally, this question was used further to evaluate the usability of the systems in our study.

For the survey analysis, we used the methodology described in Egelman and Peer [19], i.e.; we followed the scale development procedure the authors utilized. We constructed our questions in either a free response format, a multiple choice selection format, or a 5-point Likert scale to gauge responses. We also implemented attention-check questions in order to filter out careless responses. We collected 39 responses throughout the study focusing on the University students. One of the responses was discarded because the response was incomplete, and the attention check questions were incorrectly answered. We ensured that the attention check questions were answered correctly for the rest of the responses. As a result, all the participants answered these questions correctly, leaving us with a total of 38 responses to perform data analysis.

We used thematic analysis to group our data and report our findings. This was done based on individual questions. For the free-response questions, the two primary researchers of this work created a codebook to categorize the types of responses and then place each response in the respective category. After performing the initial analysis on individual questions with an inter-coder reliability score of 89.6%, we grouped all of our information through discussion. The three major themes included are; ease of use (efficiency) and reliability (error tolerance), and biometric authentication security regarding multi-factor authentication.

4 RESULTS

Our survey collected demographic data on the subjects summarized in table 1.

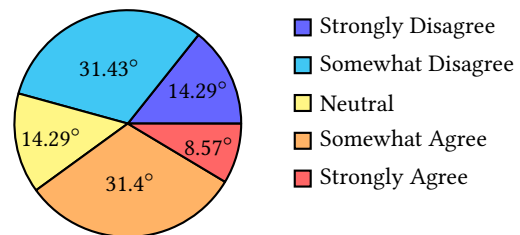
4.1 Aggregated Behavior

We collected information based on behavior to identify if our subjects have engaged with biometric authentication and, if so, what specific technologies they have used. This gave us an idea of the overall adoption that these systems have. Our subjects almost equally used Touch ID or Face ID to authenticate into their devices, with very few respondents using Voice ID. We will elaborate further in the discussion section on the differences in responses between Touch ID and Face ID. The preference distribution was Touch ID (50%), Face ID (45.45%), and Voice ID (4.55%).

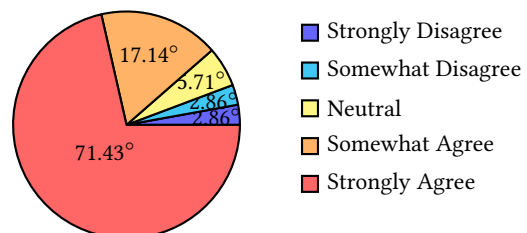
We also wanted to ask our subjects if they once used biometric authentication and stopped for any reason(s). A thorough inspection of the survey replies shows that only one of our respondents had stopped the use of biometric authentication, and this was due to their phone being repaired (replacing Touch ID reader) and their phone no longer letting them use Touch ID, but not by their own choice. The individual later got a new phone and resumed usage of Touch ID.

4.2 Aggregated Opinions

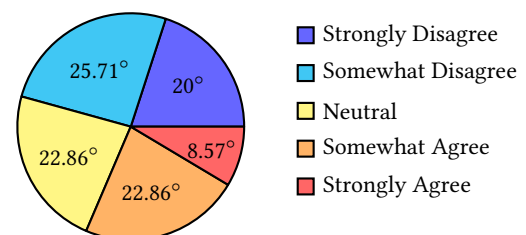
These questions sample the subjects' general opinions on phone security and usability. For example, we found a mixed response when we asked about the annoyance of using fingerprint-based biometric authentication to unlock the phone.



These numbers show that the aggregate opinion of this sample is roughly neutral concerning being annoyed by unlocking their phone. One-third of respondents indicate that using fingerprints upsets them, and a similar number of respondents indicate that they are not upset about using fingerprints. This informs us that using fingerprints does not necessarily annoy users in general. Similarly, we compared biometric authentication's security and access control, where we found a positive response. It was clear that many respondents agreed that biometric authentication is a more secure mode of authentication; over one-third of the respondents attested to it.



We also wanted to see how users react to the traditional authentication techniques of password/pattern/PIN-based authentication. Users agreed when asked if they were concerned that someone might be observing them unlocking passwords/patterns/PINs to access their phones later. This informs that many of our participants understood the limitations of password/pattern/PIN and its susceptibility to shoulder surfing.



4.3 Open response Analysis

We asked the subjects eight different open-response questions surrounding the topic of biometric authentication. We generated 35 different codes based on responses to those questions. These open-response questions were created in such a way as to determine general subject opinion on biometric authentication and to identify any potential concerns subjects have with respect to the security and usability of biometric authentication systems. The thematic analysis results will be elaborated on in the discussion section.

We found a generic security concern for FaceIDs, particularly for third-party access. One of the participants mentioned:

	<18	18-24	25-30	31-40	41-50	51-60		Female	Male	Trans	Non-binary
Subject Age	1	17	6	3	4	2	Subject Gender	13	15	1	1
				Asian	Black	Pacific Islander	White	Hispanic	Arabic	Others	
Subject Racial Identification				9.09%	3.03%	3.03%	72.72%	9.09%	3.03%	0	
				HS	Diploma	Vocational Training	UGrad	Grad	Associates		
Educational Background				27.7%	5.5%	2.7%	30.6%	30.6%	2.7%		

Table 1: Participant Demographics

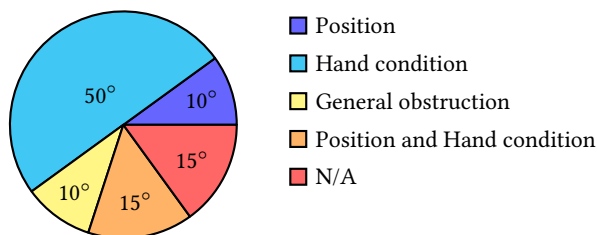
"I have to add about the Face ID. My friend can open her sister's smartphone with her face. I saw that with myself. This means the Face ID is inaccurate because it accepts the ID for the two sister's faces in one device. The sister's device is the iPhone 13 Pro."

However, many participants preferred biometric authentication in comparison to traditional authentication approaches.

"I think biometrics are more secure than having just a passcode, but maybe a combination of the two would be the most secure."

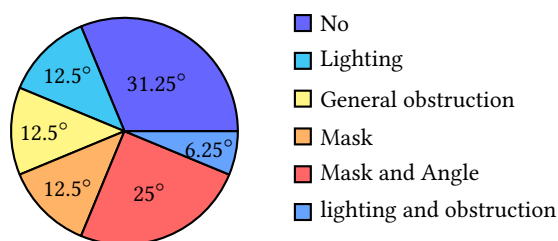
We also discussed whether MFA could be a solution and 28.5% of the participants agreed. Usability (68.18%) is the main reason for subjects to use biometrics in our study, where some of the most common reasons indicated are: *"ease of logging in, quick, do not have to remember a password, more secure."*

Half of our respondents indicated that the positioning of the Touch ID is the most prominent issue they have encountered. At the same time, a few numbers (n=10%) mentioned Hand condition and General obstruction, respectively.



We also asked: *What issues, if any, have you encountered with facial recognition in different settings (indoors, dim lighting, while walking, etc.)?* a participant mentioned:

"Masks make it difficult, and some angles do not recognize my face"



Further looking at our data, it is apparent that Touch ID has a higher rate of usability issues but fewer unique types of usability problems

than Face ID. For example, most of the problems relating to Touch ID involve different obstructions to the hands. In contrast, Face ID had a more comprehensive range of usability issues, such as lighting, angle, and obstructions. Face ID also has what one might consider a transient usability problem in the form of masks, with nearly 36% of respondents reporting usability problems related to mask-wearing.

5 DISCUSSION

In the results section, we formed a grouping based on subject practice and opinion. The following subsections combine the subset of Multiple choice and Likert questions linked to security and usability and provide thematic analysis to allocate responses to our open-response questions to those same categories.

5.1 Biometric Authentication: Usability Consideration

We found that participants will use biometric authentication as long as it is more convenient than available alternatives. Specifically, by looking at only responses that are coded as "lighting," "general obstructions," and "light and obstruction," we see that 31.25% of our subjects have usability issues with facial recognition, which can be a concern as usability is the primary aspect in biometric authentication. We found that our participants overwhelmingly think that biometric authentication mechanisms have usability issues. Subjects had more usability issues with facial recognition than fingerprint recognition. Conversely, facial recognition had a lower incidence of issues, where 31% had no problems, compared to fingerprint recognition, where only 15% had no problems. This could be due to the higher number of users who had experience with Touch ID versus Face ID. None of the responses and corresponding codes were long-term issues. For example, most complaints were tied to wet or dirty hands. None of the issues were inherently worse than or different from PIN authentication. For instance, capacitive touch screens have problems reading user input when users have wet hands or gloves.

5.2 Biometric Authentication: Security Considerations

Overwhelmingly subjects used biometric authentication at least partly because of usability, with 86% of subjects' responses coded as "security" or "security and usability." This was an open-response question; however, respondents still needed to provide reasons why they thought biometrics were more secure, indicating the belief might not be backed up by education on biometric authentication, somewhat off on experience. Subjects answered affirmatively when

asked about better ways to have stronger authentication than biometrics, with 56% of responses coded as either "yes" or "yes with multi-factor authentication." These responses lead us to state that our subjects are still generally concerned about biometric authentication security and think some enhancements could be made. For example, one participant responded,

"I think biometrics are more secure than having just a passcode, but maybe a combination of the two would be the most secure."

Participants mostly remarked on ease of use; some said it was both easy and secure. Our results indicate that security is a reason the subjects used biometric authentication, but the primary reason was the ease of use. Only 31% of responses are coded as solely "security" or "security and usability." We also found that while the subjects highly value security, there is an overall ambivalence towards specific issues surrounding the security of PIN/password/pattern authentication. Subjects in our survey generally understood the security implications of biometric authentication. However, subjects in our survey also had concerns about biometric authentication and how it related to security. As researchers, we need to recognize this because biometric authentication is not a panacea. PIN/password/passcode still has a role in securing devices and authentication schemes.

6 IMPLICATIONS

We formed the following recommendations by surveying subject behavior and prior work. Unfortunately, we also encountered subjects blindly trusting biometric authentication as "unhackable," creating a false sense of security.

6.1 Inclusive and Usable Biometric Authentication

Usability is the primary reason our subjects use biometric authentication. While they have noted some challenges in the usage, it is still vastly more usable than PIN/Password/Passcode authentication mechanisms. We investigated the two authentication mechanisms separately; we can see that Touch ID had definable usability problems very quickly for subjects, mostly to do with the skin condition of the hand/finger. Face ID had a more comprehensive arrangement of issues but fewer total issues. We have not tried to link Face ID to specific respondents to evaluate equity, but it is not out of the realm of possibility, given that our sample is skewed toward white males. The fact that usability issues were more uniform on Touch ID suggests that Touch ID might be equitable across diverse subjects. As recommended by prior researchers, we need to focus on creating a socially-inclusive digital identity, especially in vulnerable populations or people with disability; people with loss of dexterity may struggle to use Touch ID, a person with voice tremor may struggle with Voice ID, while a blind person may have Face ID not working for them [15, 20, 21, 43].

6.2 Risk Communication: False Sense of Security

Security is critical, especially when it comes to biometric data. Over 87 % of respondents agree that they "like the idea that the phone

is protected from unauthorized access." Most participants treated it as "unhackable"; however, even one of our participants noted how it was "easily hacked" by their sibling. There is a line of research where twins could hack each other's cellphones due to the similarity of their facial structure [22, 29]. The danger in biometric authentication includes the biometric data collected at every point of authentication, data must be encrypted, and the assets behind the authentication must be secured because if compromised, attackers can masquerade as a legitimate and authenticated user [31]. Everyone needs to consider going forward the opportunity to become educated on available authentication schemes; this education could be beneficial in a higher volume to ensure users are well aware of the risks associated with their chosen authentication schemes. Such risk communication to the users [2, 5, 36] is imperative to remove the false sense of security and not over-rely on biometric authentication to have a fail-safe mechanism in place.

6.3 Design Alterations

Though usability is a primary motivating factor, we must stay up-to-date with technological and social developments and changes. One such change is recognizing that masks are essential to life due to COVID-19 and other medical circumstances necessitating one. Many participants expressed concerns about not being able to use Face IDs in public transportation while using masks. Cellphones are currently making changes to adjust to that [1, 33, 37, 40]; however, removing some of the facial features to address authentication with masks creates a security vulnerability. From our findings, this subject is not alone in thinking of a higher level of security with biometric authentication versus a PIN. However, someone can see the subject's face using a Face ID and probably see it better than a Pin/Password/Passcode. The question is if they can replicate the subject's face. Thus, a new interface that introduces a design balance must be achieved.

In addition, the current design for Touch ID threatens its security; fingerprint cloning can be done with a camera and mobile application. A hacking event in China had Tencent Security's X-Lab team leader Chen Yu take a picture of a fingerprint on glass and extract the data that enabled them to create a physical version of the fingerprint with a mobile application in 20 minutes [4]. The current design of Face ID makes it susceptible to deepfakes, a maturing technology. Recently, hackers purchased high-quality facial images from a black market and used an app to create deepfakes to trick a tax invoice system [41].

7 FUTURE WORK AND LIMITATIONS

We acknowledge that for a survey-based study, 38 responses seem low. However, we mainly focused on high-tech university students to evaluate the impact of biometric authentication on a tech-savvy user base. Furthermore, we plan to extend this study to deploy it in crowdsourcing platforms such as Prolific and Amazon MTurk to get a quantitative overview [28]. Lastly, in the investigation into other forms of biometric authentication, we had one subject who had experience with Voice ID. We plan to investigate further while exploring the algorithmic biases; any inherent bias towards people speaking English as a second language or a regional dialect would be interesting extended work.

8 CONCLUSION

Biometric authentication has gained an increased level of adoption based on its ease of usage in recent times; however, the security challenge therein has yet to be discovered by users. In this work, we investigated the usage of biometric authentication through a survey-based study collecting data from 38 participants in a University setting. Our results showed that, overall, participants favored biometric authentication due to the usability of the different mechanisms. Additionally, subjects had some concerns about the security of specific biometric technologies but did not have any comments on them. Our results suggest insufficient security risk communication of using biometric authentication and that it is marketed solely for its ease of use. However, subjects were interested in using biometric authentication as an additional factor in multi-factor authentication schemes to enhance their security.

Furthermore, we found that participants found Touch ID and Face ID were transient, i.e., things that will resolve themselves in time: wet hands, masks, lighting, and others. Participants also demonstrated somewhat of a naive attitude toward biometric authentication that needs to be curbed. In other words, they were using it without much education on its associated risks and trusted that it was secure. We report the results and provide actionable recommendations for improved biometric authentication user adoption.

ACKNOWLEDGMENTS

We would also like to acknowledge the Inclusive Security and Privacy-focused Innovative Research in Information Technology (InSPIRIT) Lab at the University of Denver, where this research was conducted. Any opinions, findings, conclusions, or recommendations expressed in this material are solely those of the authors.

REFERENCES

- [1] Aqeel Anwar and Arijit Raychowdhury. 2020. Masked face recognition for secure authentication. *arXiv preprint arXiv:2008.11104* 1 (2020), 1–8.
- [2] Gökhan Bal, Kai Rannenberg, and Jason I Hong. 2015. Styx: Privacy risk communication for the Android smartphone platform based on apps' data-access behavior patterns. *Computers & Security* 53 (2015), 187–202.
- [3] Rasekhar Bhagavatula, Blase Ur, Kevin Iacovino, Su Mon Kywe, and Lorrie Faith Cranor. 2015. Biometric authentication on iPhone and Android: Usability, perceptions, and influences on adoption. *USEC '15: Workshop on Usable Security 2* (Feb 2015).
- [4] Aleksandar Branković. 2020. How biometrics provide an amazing UX, but fail too often. <https://www.ipification.com/blog/how-biometrics-provide-an-amazing-ux-but-fail-too-often/>
- [5] Arslan Brömmle. 2006. A Risk Analysis Approach for Biometric Authentication Technology. *Int. J. Netw. Secur.* 2, 1 (2006), 52–63.
- [6] Xavier Bultel, Jannik Dreier, Matthieu Giraud, Marie Izaute, Timothée Kheyrikhah, Pascal Lafourcade, Dounia Lakhzoum, Vincent Marlin, and Ladislav Motá. 2018. Security analysis and psychological study of authentication methods with PIN codes. In *2018 12th International Conference on Research Challenges in Information Science (RCIS)*. IEEE, IEEE, Nantes, France, 1–11.
- [7] Cynthia M. Cook, John J. Howard, Yevgeniy B. Sirotin, Jerry L. Tipton, and Arun R. Vemury. 2019. Demographic Effects in Facial Recognition and Their Dependence on Image Acquisition: An Evaluation of Eleven Commercial Systems. *IEEE Transactions on Biometrics, Behavior, and Identity Science* 1, 1 (2019), 32–41. <https://doi.org/10.1109/TBIOM.2019.2897801>
- [8] Neha Dahiya and Chander Kant. 2012. Biometrics Security Concerns. In *2012 Second International Conference on Advanced Computing Communication Technologies*. IEEE, Haryana, India, 297–302. <https://doi.org/10.1109/ACCT.2012.36>
- [9] Sanchari Das, Andrew Dingman, and L Jean Camp. 2018. Why Johnny doesn't use two factor a two-phase usability study of the FIDO U2F security key. In *International Conference on Financial Cryptography and Data Security*. Springer, Nieuwpoort, Curaçao, 160–179.
- [10] Sanchari Das, Robert S Gutzwiller, Rod D Roscoe, Prashanth Rajivan, Yang Wang, L Jean Camp, and Roberto Hoyle. 2020. Humans and technology for inclusive privacy and security. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. SAGE Publications Sage CA: Los Angeles, CA, HFES, Online, 461–464.
- [11] Sanchari Das, Andrew Kim, Ben Jelen, Lesa Huber, and L Jean Camp. 2020. Non-inclusive online security: older adults' experience with two-factor authentication. In *Proceedings of the 54th Hawaii International Conference on System Sciences*. Hawaii International Conference on System Sciences, Honolulu, Hawaii.
- [12] Sanchari Das, Andrew Kim, Ben Jelen, Joshua Streiff, L. Jean Camp, and Lesa Huber. 2019. Towards Implementing Inclusive Authentication Technologies for Older Adults. In *Who Are You?! Adventures in Authentication Workshop (WAY '19)*. WAY, Santa Clara, California, USA, 1–5.
- [13] Sanchari Das, Andrew Kim, Ben Jelen, Joshua Streiff, L Jean Camp, and Lesa Huber. 2020. Why Don't Older Adults Adopt Two-Factor Authentication?. In *Proceedings of the 2020 SIGCHI Workshop on Designing Interactions for the Aging Populations-Addressing Global Challenges*. ACM, Virtual.
- [14] Sanchari Das, Andrew Kim, Shrirang Mare, Joshua Streiff, and L Jean Camp. 2019. Security mandates are pervasive: An inter-school study on analyzing user authentication behavior. In *2019 IEEE 5th International Conference on Collaboration and Internet Computing (CIC)*. IEEE, IEEE, Los Angeles, USA, 306–313.
- [15] Sauvik Das, Gierad Laput, Chris Harrison, and Jason I Hong. 2017. Thumbprint: Socially-inclusive local group authentication through shared secret knocks. In *Proceedings of the 2017 chi conference on human factors in computing systems*. ACM, Denver, USA, 3764–3774.
- [16] Sanchari Das, Gianpaolo Russo, Andrew C Dingman, Jayati Dev, Olivia Kenny, and L Jean Camp. 2018. A qualitative study on usability and acceptability of Yubico security key. In *Proceedings of the 7th Workshop on Socio-Technical Aspects in Security and Trust*. ACM, Orlando, Florida, 28–39.
- [17] Sanchari Das, Bingxing Wang, Zachary Tingle, and L Jean Camp. 2019. Evaluating User Perception of Multi-Factor Authentication: A Systematic Review. In *Proceedings of the Thirteenth International Symposium on Human Aspects of Information Security & Assurance (HAISA 2019)*. arxiv, Nicosia, Cyprus, 10 pages.
- [18] Reyhan Düzgün, Naheem Noah, Peter Mayer, Sanchari Das, and Melanie Volkamer. 2022. SoK: A Systematic Literature Review of Knowledge-Based Authentication on Augmented Reality Head-Mounted Displays. In *Proceedings of the 17th International Conference on Availability, Reliability and Security*. ACM, Vienna, Austria, 1–12.
- [19] Serge Egelman and Eyal Peer. 2015. Scaling the Security Wall: Developing a Security Behavior Intentions Scale (SeBIS). In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI '15). Association for Computing Machinery, New York, NY, USA, 2873–2882. <https://doi.org/10.1145/2702123.2702249>
- [20] Lothar Fritsch, Kristin Skeide Fuglerud, and Ivar Solheim. 2010. Towards inclusive identity management. *Identity in the Information Society* 3, 3 (2010), 515–538.
- [21] Alan Gelb and Anit Mukherjee. 2019. Building on digital ID for inclusive services: lessons from India. *CGD NOTES* 2020 (2019), 1–10.
- [22] Vitaly Gnatyuk and Alena Moskalenko. 2020. Mobile Twin Recognition. In *2020 IEEE International Joint Conference on Biometrics (IJCB)*. IEEE, IEEE, Texas, USA, 1–9.
- [23] Merideth Going and J. D. Read. 1974. Effects of Uniqueness, Sex of Subject, and Sex of Photograph on Facial Recognition. *Perceptual and Motor Skills* 39, 1 (1974), 109–110. <https://doi.org/10.2466/pms.1974.39.1.109>
- [24] Javier Guerra-Casanova, Belén Ríos-Sánchez, Miguel Viana-Matesanz, Gonzalo Bailador, Carmen Sánchez-Ávila, and María José Melcón De Giles. 2016. Comfort and Security Perception of Biometrics in Mobile Phones with Widespread Sensors. In *2016 IEEE 35th Symposium on Reliable Distributed Systems (SRDSW)*. IEEE, Budapest, Hungary, 13–18. <https://doi.org/10.1109/SRDSW.2016.13>
- [25] Marian Harbach, Emanuel Von Zezschwitz, Andreas Fichtner, Alexander De Luca, and Matthew Smith. 2014. It's a Hard Lock Life: A Field Study of Smartphone (Un)Locking Behavior and Risk Perception. In *Proceedings of the Tenth USENIX Conference on Usable Privacy and Security* (Menlo Park, CA) (SOUPS '14). USENIX Association, USA, 213–230.
- [26] Marian Harbach, Emanuel Zezschwitz, Andreas Fichtner, Alexander De Luca, and Matthew Smith. 2014. It's a Hard Lock Life: A Field Study of Smartphone (Un)Locking Behavior and Risk Perception. In *SOUPS*. ACM, California, USA.
- [27] Rosa R. Heckle, Andrew S. Patrick, and Ant Ozok. 2007. Perception and Acceptance of Fingerprint Biometric Technology. In *Proceedings of the 3rd Symposium on Usable Privacy and Security* (Pittsburgh, Pennsylvania, USA) (SOUPS '07). Association for Computing Machinery, New York, NY, USA, 153–154. <https://doi.org/10.1145/1280680.1280704>
- [28] Matthias Hirth, Tobias Hofffeld, and Phuoc Tran-Gia. 2011. Anatomy of a crowdsourcing platform-using the example of microworkers. com. In *2011 Fifth international conference on innovative mobile and internet services in ubiquitous computing*. IEEE, IEEE, Korean Bible University (KBU), Seoul, Korea, 322–329.
- [29] S Jayanthi, S Sivakumar, and R Murugesan. 2017. Empathize an Identical Twins using Face Recognition. *International Journal of Computational and Applied Mathematics* 12, 1 (2017), 2017.

- [30] John M Jones, Reyhan Duezguen, Peter Mayer, Melanie Volkamer, and Sanchari Das. 2021. A literature review on virtual reality authentication. In *International Symposium on Human Aspects of Information Security and Assurance*. Springer, Springer, Virtual, 189–198.
- [31] Vishruti Kakkad, Meshwa Patel, and Manan Shah. 2019. Biometric authentication and image encryption for image security in cloud framework. *Multiscale and Multidisciplinary Modeling, Experiments and Design* 2, 4 (2019), 233–248.
- [32] Atul N. Kataria, Dipak M. Adhyaru, Ankit K. Sharma, and Tanish H. Zaveri. 2013. A survey of automated biometric authentication techniques. In *2013 Nirma University International Conference on Engineering (NUICONE)*. IEEE, Ahmedabad, India, 1–6. <https://doi.org/10.1109/NUICONE.2013.6780190>
- [33] Seong Tae Kim and Yong Man Ro. 2018. Attended relation feature representation of facial dynamics for facial authentication. *IEEE Transactions on Information Forensics and Security* 14, 7 (2018), 1768–1778.
- [34] Ritajit Majumdar and Sanchari Das. 2021. Sok: An evaluation of quantum authentication through systematic literature review. In *Proceedings of the Workshop on Usable Security and Privacy (USEC)*. NDSS, Auckland, New Zealand - Virtual, 10 pages.
- [35] Carsten Maple and Peter Norrington. 2006. The Usability and Practicality of Biometric Authentication in the Workplace. In *Proceedings of the First International Conference on Availability, Reliability and Security*. ACM, Vienna, Austria, 958–964.
- [36] Giovanni L Masala, Pietro Ruii, and Enrico Grosso. 2018. Biometric authentication and data security in cloud computing. In *Computer and network security essentials*. Springer, Athens, 337–353.
- [37] S Naveen, R Shihana Fathima, and RS Moni. 2016. Face recognition and authentication using LBP and BSIF mask detection and elimination. In *2016 International Conference on Communication Systems and Networks (ComNet)*. IEEE, IEEE, Thiruvananthapuram, India, 99–102.
- [38] L. O’Gorman. 2003. Comparing passwords, tokens, and biometrics for user authentication. *Proc. IEEE* 91, 12 (2003), 2021–2040. <https://doi.org/10.1109/JPROC.2003.819611>
- [39] Prashanth Rajivan, Pablo Moriano, Timothy Kelley, and L Jean Camp. 2017. Factors in an end user security expertise instrument. *Information & Computer Security* 25 (2017), 2.
- [40] Navya Saxena and Devina Varshney. 2021. Smart Home Security Solutions using Facial Authentication and Speaker Recognition through Artificial Neural Networks. *International Journal of Cognitive Computing in Engineering* 2 (2021), 154–164.
- [41] Help Net Security. 2021. Combating deepfakes: How we can future-proof our biometric identities. <https://www.helpnetsecurity.com/2021/07/20/combating-deepfakes/>
- [42] Flynn Wolf, Ravi Kuber, and Adam J. Aviv. 2019. "Pretty Close to a Must-Have": Balancing Usability Desire and Security Concern in Biometric Adoption. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300381>
- [43] Rita Wong, Norman Poh, Josef Kittler, and David Frohlich. 2010. Towards inclusive design in mobile biometry. In *3rd International Conference on Human System Interaction*. IEEE, IEEE, Rzeszow, Poland, 267–274.

A SURVEY QUESTIONNAIRE

A.1 Pre-Screening Questions

- (1) What is your age?
 - Less than 18
 - 18-24 years
 - 25-30 years
 - 31-40 years
 - 41-50 years
 - 51-60 years
 - 61-70 years
 - More than 70 years

A.2 Demographics Questions

- (1) What ethnicity do you identify with? (Please select all that apply)
 - American Indian or Native American
 - Asian
 - Black or African American
 - Native Hawaiian or Other Pacific Islander
 - White
 - Hispanic
 - Other. Please Specify
 - Do not Wish to Specify
- (2) Which gender do you identify with the most?
 - Female
 - Male
 - Transgender
 - Non-binary
 - A gender not listed here
 - Do not Wish to Specify
- (3) What is the highest level of education you have completed? (If currently enrolled, highest degree received.)
 - Less than high school
 - High school graduate
 - Diploma
 - Vocational training
 - Bachelors degree program
 - Masters degree program
 - Professional degree
 - Doctorate
 - Other, Please Specify

A.3 Cyber Background Questions

- (1) Please rank the following statement on a scale between Strongly Disagree to Strongly Agree: "I often ask others for help with the computer."
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (2) Please rank the following statement on a scale between Strongly Disagree to Strongly Agree: "Others often ask me for help with the computer."
 - Strongly Disagree
 - Somewhat Disagree

- Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (3) Which of the following things have you done?
 - Designed a website
 - Registered a domain name
 - Used SSH
 - Configured a firewall
 - Created a database
 - Installed a computer program
 - Written a computer program
 - None of the above
 - (4) Do you have an up-to-date virus scanner on your computer?
 - Yes
 - No
 - (5) Have you taken or taught a course on computer security?
 - Yes
 - No
 - (6) Is computer security one of your main job responsibilities?
 - Yes
 - No
 - (7) Do you have a security cognate or do you plan on security being a feature of your job role?
 - Yes
 - No
 - (8) To your knowledge, have you ever been hacked or compromised by attackers?
 - Yes
 - No
 - I don't know
 - (9) Have you ever suffered data loss for any reason? (ex: Hacking, data corruption, hard drive failure)
 - Yes
 - No
 - I don't know
 - (10) Please indicate how well you agree or disagree with the following statements: "I can control the risk of my online account being taken over or being phased."
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
 - (11) Please indicate how well you agree or disagree with the following statements: "The harm of being locked out of my account would be immediate."
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
 - (12) Please indicate how well you agree or disagree with the following statements: "If I am exposed to online risk, I can mitigate the harm."
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree

- Somewhat Agree
 - Strongly Agree
- (13) Please indicate how well you agree or disagree with the following statements: "Experts understand phishing and know how to protect me from it."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (14) Please indicate how well you agree or disagree with the following statements: "I understand phishing and know how to protect myself."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (15) Please indicate how well you agree or disagree with the following statements: "Phishing is widespread and affects many people, and not only for target individuals."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (16) Please indicate how well you agree or disagree with the following statements: "Phishing is everywhere. It is a constant exposure."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (17) Please indicate how well you agree or disagree with the following statements: "Phishing is a new kind of risk."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree

A.4 General Biometric Usability Questions

- (1) Have you ever used biometric authentication measures to secure your device(s)?
 - Yes
 - No
- (2) Are you currently using biometric measures to secure your device(s)?
 - Yes
 - No
- (3) What types of biometric authentication have you used, or are currently using?
 - Face ID
 - Voice ID
 - Eyes/Iris ID
 - Other (please specify)

- (4) If you are currently using biometrics to secure your device(s), why do you use it?
- (5) If you are no longer using biometrics to secure your device(s) but have in the past, why are you no longer using it?
- (6) If you have never used biometrics to secure your device(s), what are the reasons why? What would if anything, make you decide to use such a scheme to secure your device(s)?
- (7) If you have never used biometrics to secure your device(s), what are the reasons why? What would if anything, make you decide to use such a scheme to secure your device(s)?
 - High security and assurance - it provides the answers to "something a person has and is" and helps verify identity
 - User experience - convenient and fast
 - Non-transferrable - everyone has access to a unique set of biometrics
 - Spoof proof - biometrics are hard to fake or steal
 - Other (please specify)
- (8) What disadvantages do you find biometric authentication to have compared to other schemes of security?
 - Costs - significant investment needed in biometrics for security
 - Data breaches - biometric databases can still be hacked
 - Tracking and data - biometric devices can limit privacy for users
 - Bias - machine learning and algorithms must be very advanced to minimize biometric demographic bias
 - False positives and inaccuracy - false rejects and false accepts can still occur preventing select users from accessing systems
 - Other (please specify)
- (9) What issues, if any, have you encountered with facial recognition in different settings (indoors, dark lighting, while walking, etc)?
- (10) What issues, if any, have you encountered with Touch ID in different settings (dirty hands, while walking, etc.)?
- (11) What issues, if any, have you encountered with voice ID?
- (12) What issues, if any, have you encountered with Eye/Iris ID?

A.5 General Device Security Questions

- (1) Please rate the following statements concerning your lock screen: "Unlocking my phone is annoying sometimes."
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (2) Please rate the following statements concerning your lock screen: "I like the idea that my phone is protected from unauthorized access."
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (3) Please rate the following statements concerning your lock screen: "It is difficult to unlock my phone."

- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (4) Please rate the following statements concerning your lock screen: "I wish there was an easier way of unlocking my phone."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (5) Please rate the following statements concerning your lock screen: "Unlocking my phone is easy."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (6) I am concerned that someone might be observing my unlocking password/pattern/PIN in order to access my phone at a later time."
- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- (7) Do you sometimes take additional measures to protect your smartphone in particular situations? Choose all that apply.
- I leave my phone in a safe place before going somewhere.
 - I conceal my smartphone in my clothes or in a bag.
 - I enable a lock screen for this situation or choose a harder PIN/password/pattern.
 - Other
- (8) If you think someone is able to see the screen of your phone, do you sometimes take additional measures to protect your smartphone? Choose all that apply.
- I cover my smartphone while entering my PIN or pattern.
 - I wait a moment before entering my PIN or pattern.
 - I turn around before entering my PIN or pattern.
 - I tilt my screen away before entering my PIN or pattern.
 - I change my PIN/password/pattern after someone could have seen my screen.
 - Other